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GRINS FOUNDATION

SCI-WEB

Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being

DELIVERABLE 2.1

Dataset of relevant variables for multivariate well-being at provincial and detailed scales

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Executive summary

This report presents the list of relevant variables developed to support the analysis of local inequalities and territorial well-being at geographic scales more detailed than NUTS2, as part of SCI-WEB project. This work is carried out within the broader framework of the GRINS project and specifically aligns with the objectives of Spoke 8.

The datasets are based on the key concepts outlined in the first Deliverable of the SCI-WEB project and are conceptually and methodologically connected to ISTAT's *Benessere Equo e Sostenibile* (BES) and its territorial extension, BES-T. As detailed in Table 1, all variables are linked to the 11 BES-T domains, with an enhanced focus on education, innovation, and migration (included in the social relationship domain; among others, Blumentstock et al. 2025).

In an era marked by recurring crises that challenge the resilience of local communities, it is vital to explore well-being and inequality in depth, without overlooking their environmental and social roots. These perspectives are essential for raising awareness and driving a cultural and institutional shift toward multidimensional policy frameworks and place-based strategies. In this context, SCI-WEB offers tools designed to assess different pillars of sustainability undertaken by enterprises and public administrations at granular geographical scales.

Table 1- Domains considered for both Benessere Equo e Sostenibile report at the national level (BES) and the territorial Benessere Equo e Sostenibile (BES-T).

Domains	***BES***	***Territorial BES And SCI-WEB***
Environment	✓	✓
Economic well-being	✓	✓
Innovation, research and creativity	✓	✓
Education and training	✓	✓
Work and life balance	✓	✓
Landscape and cultural heritage	✓	✓
Politics and institutions	✓	✓
Quality of servicesv	✓	✓
Social relationships	✓	✓
Health	✓	✓
Safety	✓	✓
Subjective well-being	✓	
Domains considered	**12**	**11**

Accordingly, careful consideration has been devoted to key variables related to pollution, climate change adaptation, and the implementation of more **sustainable models**. These variables are

designed to provide localized insights that inform everyday challenges and support sustainability-oriented decision-making at the territorial level (Cellini and Torrisi, 2014). Cultural dimensions are non-secondary, given the importance of managing these activities within a socially and environmentally sustainable framework. Particular attention has been placed on the presence and role of historical parks, farms, and tourism activities (Bernini et al. 2021). Moreover, the datasets support the idea of considering relevant **gender differences** (especially over the labour and education dimension), aligning with the thematic priorities of Spoke 8 and the broader GRINS project.

The main idea is that all these dimensions are crucial to the scientific objective of GRINS and – at a more general level – for the sustainable growth objective outlined within the National Recovery and Resilience Plan. Indeed, only a multidimensional approach to territorial development can effectively support more just, socially inclusive, and environmentally sustainable outcome. The variables presented in this Deliverable follow this perspective. Variable selection boosts its adherence to the Spoke 8, including the focus given to environment, culture, gender differences and quality of services (e.g., waste management indicators are collected at different levels).

To expand the analytical potential of the dataset and meet the needs of a diverse range of users, we have integrated variables from **a wide array of sources** beyond the BES framework. These include the Annual Regional Database of the European Commission (ARDECO), European Environmental Agency (EEA), Database of the "*Autorità Garante delle Telecomunicazioni*" (AGCOM), ISPRA (Italian Institute for Environmental Protection and Research), the European Union's Earth Observation Programme, the Italian Ministry of Economic and Finance, the Italian Ministry of Culture, The Italian Ministry of Education, Eurostat, OECD, European Data Journalism Network and the Statistical Atlas by ISTAT, among others.

In this report, each variable is accompanied by **metadata** that includes general information on data integration strategies and relevant methodological references. Further methodological details - concerning the handling of spatial missing data and the disaggregation techniques to obtain data at finer geographic scales are thoroughly documented in *Deliverable D2.2: Report on Explorative Spatial Analysis of Relevant Variables of Well-being*. The Deliverable D.2.2 complements the present report and is being released concurrently.

The dataset is structured in two main parts. The first part includes the **52 indicators at the NUTS3 level** (i.e., provincial level), forming a balanced dataset over the period 2014–2021. At least four indicators are provided for each of the 11 BES-T domains. The selected time frame aims to maximize the availability and continuity of territorial indicators, giving a picture of a pre-pandemic and post the COVID-19 outbreak in 2020. The second part focuses on data at the **municipal level**. Due to limitations in data availability, this section concentrates on two reference years: 2018 and 2021. **It includes 25 different variables** with full spatial coverage, ensuring representation for all eleven BES-T domains, with at least one variable per domain.

Ensuring a wide range of indicators at municipal level is particularly challenging. Nevertheless, through the use of proxy measures and alternative indicators, we were able to

represent all eleven BES-T domains and deliver a multidimensional dataset at this level.The availability of data at this finer resolution is particular valuable for revealing **intra-regional disparities** and spatial inequalities in well-being.

Overall, a particular emphasis has been placed on capturing the multidimensional nature of well-being, with focused attention on environmental sustainability, cultural dimensions, and gender inequalities. These aspects are closely aligned with the goals of the GRINS and AMELIA platform, and are particularly central to the mission of Spoke 8.

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1. Introduction

This report is primarily devoted to the description of the variables included in the SCI-WEB datasets. The accompanying metadata provides comprehensive information on 52 variables at the NUTS3/provincial level and 25 variables at LAU (Local Administrative Units according to the Eurostat definition) or municipal level in Italy. For each variable, the metadata includes the variable name, dataset codes, domain classification, descriptive notes, data sources, reference links, spatial and temporal coverage, and the geographical classification system used.

Variable definitions are given according to the official source and, whenever possible, links to English versions of those definitions have been reported in the metadata. Table 2 presents the full list of domains and their acronyms, alongside a summary of the spatial and temporal coverage of the datasets.

Table 2- Domains considered in SCI-WEB and the acronyms used encoding variables in the datasets release of the Deliverable 2.1.

Domains	***Acronym at NUTS3 level***	***Acronym at municipality level***
Environment	A	a
Economic well-being	BE	be
Innovation, research and creativity	I	i
Education and training	IS	is
Work and life balance	L	l
Landscape and cultural heritage	PA	pa
Politics and institutions	PO	po
Quality of services	Q	q
Social relationships	R	r
Health	H	h
Safety	S	s
Years covered	2014-2021 (8 years)	2018 and 2021 (2 years)
Number of spatial units	107 for all years	7960 (2018) 7904 (2021)

As shown in Table 2, the dataset at the NUTS3 level is based on the current administrative configuration of 107 spatial units. Where needed, data were adjusted or integrated to obtain a balanced panel dataset, reconciling it with the previous structure of 110 provinces. At the municipal level, recent changes in administrative boundaries posed additional challenges. Accordingly, 7,960 municipalities were considered for the year 2018 and 7,904 for 2021, based on

ISTAT classifications (ISTAT 2024; 2025). Table 3 summarize the number of variables included in each dataset and each domain.

Table 3- Number of variables considered in SCI-WEB datasets and specification of the number of variables for each domain.

Domains	***Number of variables per domain at NUTS3 level***	***Number of variables per domain at municipality level (2018 and 2021)***
Environment	5	4
Economic well-being	5	3
Innovation, research and creativity	5	2
Education and training	5	2
Work and life balance	5	3
Landscape and cultural heritage	4	2
Politics and institutions	4	2
Quality of services	5	3
Social relationships	4	2
Health	5	1
Safety	5	1
Number of variables	**52**	**25**

Metadata are provided for each variable in the datasets and serve as essential guidance for potential users. They summarize key aspects of data integration, including the treatment of missing data and the disaggregation methods used to represent variables at geographic levels below NUTS2. Detailed descriptions of the disaggregation and imputation techniques used to address missing data are available in Deliverable D2.2 of the SCI-WEB project.

It is relevant to highlight that all data included in the dataset originate from official statistics. This ensures compliance with data protection standards, including those related to informed consent. Only aggregated data are used in SCI-WEB; no individual-level data were collected, processed, or surveyed. As a result, no ethic issues have been identified according to the official EU Grants guide on "*How to Complete a Self-Assessment*"[1].

The dataset meets the specifications outlined by the SCI-WEB project proposal and is intended for use within the AMELIA platform under the GRINS Project[2], Spoke 8, and according to the *Findable, Accessible, Interoperable, and Reproducible* (FAIR) principle.

[1] *The guide is available the the following link: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/common/guidance/how-to-complete-your-ethics-self-assessment_en.pdf*

[2] https://grins.it

To support usability, this report includes links and references to help readers and users navigate across the various domains and variables, both at the provincial (NUTS3) and municipal (LAU) levels.

2. Metadata At Provincial Level

These metadata refer to 52 indicators at the NUTS3 level (i.e., provinces) covering the years from 2014 to 2021. They cover 107 statistical units, based on the definition of administrative boundaries provided by ISTAT (ISTAT, 2024).

The domains considered are the following:

- Environment (A)
- Economic well-being (BE)
- Innovation, research and creativity (I)
- Education and training (IS)
- Work and life balance (L)
- Landscape and cultural heritage (PA)
- Politics and istitutions (PO)
- Quality of services (Q)
- Social relationships (R)
- Health (H)
- Safety (S)

ENVIRONMENT (A)

The indicators of the Environment domain at provincial level aim at monitoring the evolution of citizens' well-being with respect to the natural environment in which they live.

The variables considered for this domain are:

- Urban Green (A_1)
- Consecutive Dry Days (A_2)
- Municipal Waste Generated (A_3)
- Air Pollution (A_4)
- Electricity from renewable sources (A_5)

Homepage

Urban Green (A_1)

Name: Urban Green

Code: A_1

Domain: Environment (A)

Description: Square meters of urban parks and gardens per inhabitants in provincial capital Municipalities.

Source: Istat/BES, Survey on urban environmental data

Reference: https://www.istat.it/wp-content/uploads/2024/11/10.pdf

https://www.istat.it/scheda-qualita/dati-ambientali-nelle-citta/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Consecutive Dry Days (A_2)

Name: Consecutive dry days

Code: A_2

Domain: Environment (A)

Description: Maximum number of consecutive dry days per year with daily precipitation amount of less than 1 mm.

Source: Istat/BES, Copernicus - European Union's Earth Observation Programme - Gridded dataset of climate reanalysis, ERA5 hourly data.

Reference: https://www.istat.it/wp-content/uploads/2024/11/10.pdf

https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Municipal Waste Generated (A_3)

Name: Municipal waste generated

Code: A_3

Domain: Environment (A)

Description: Municipal waste generated per capita (in kgs).

Source: Istat/BES, Processing of data from ISPRA.

Reference: https://www.istat.it/wp-content/uploads/2024/11/10.pdf

https://www.isprambiente.gov.it/en/databases/data-base-collection/waste

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Air Pollution (A_4)

Name: Air pollution

Code: A_4

Domain: Environment (A)

Description: The concentration average in µg/m³ for fine particulate matter (PM2.5) refers to the counterfactual concentrations used according to the World Health Organization 2021 Air Quality Guidelines (AQG). The reference value for health, as defined by the WHO, is 10 µg/m³.

Source: EEA - European Environment Agency

Reference: https://sdi.eea.europa.eu/catalogue/datahub/api/records/258fa83c-dec6-4d88-b1fb-959f2d90008f/formatters/xsl-view?output=pdf&language=eng&approved=true

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Electricity From Renewable Sources (A_5)

Name: Electricity from renewable sources

Code: A_5

Domain: Environment (A)

Description: Percentage of energy consumption provided by renewable sources on gross electricity consumption. The indicator is calculated as the ratio between the gross electricity production from renewable sources (actual, non-normalized) and the gross domestic consumption of electricity (e.g., the gross production of electricity, including pumping, plus trade balance).

Source: Istat/BES, Terna S.p.A., Annual statistics of electricity production and consumption in Italy.

Reference: https://www.istat.it/wp-content/uploads/2024/11/10.pdf

https://dati.terna.it/en/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: For missing units Bayesian Imputation Method (Panzera et al. 2016) has been used as explained in the SCI-WEB Deliverable D2.2.

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

ECONOMIC WELL-BEING (BE)

The indicators for the Economic Well-being domain at provincial aim to monitor the evolution of the economic well-being of citizens in the 107 provinces of Italy.

The variables considered for this domain are:

- GDP per capita at current price (BE_1)
- Income (Total income tax returns – IRPEF) (BE_2)
- Retirees with low pension income (BE_3)
- Default rate of bank loans to households (BE_4)
- Nominal labour productivity per hour worked (BE_5)

Homepage

GDP Per Capita At Current Prices (BE_1)

Name: GDP per capita at current prices

Code: BE_1

Domain: Economic well-being (BE)

Description: This indicator represents the nominal Gross Domestic Product per capita. Ratio between the gross disposable income of consumer families and the total number of residents (current prices). Nominal GDP per capita is calculated by dividing GDP at current prices by annual average population. The variable is expressed in euros.

Source: JRC / REGIO - ARDECO

Reference: https://urban.jrc.ec.europa.eu/dataset/ARDECO-SUVGDP/metadata?lng=en

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Income (Total income tax returns – IRPEF) (BE_2)

Name: Income (Total income tax returns - IRPEF)

Code: BE_2

Domain: Economic well-being (BE)

Description: The personal income tax (*Irpef*) is the tax payable by individuals for the possession of the following income: land, i.e. buildings and land; capital; employment (including income assimilated to that of employment and pension income); self-employment; businesses; various (listed in article 67 of the Consolidated Income Tax Act). Natural persons resident and non-resident in the territory of the State are taxable subjects by Irpef. Income (total income tax return - IRPEF) is calculated by dividing the "total income tax return - IRPEF" by the average annual number of taxpayers per province. The variable is expressed in euros.

Source: Revenue Agency, Statistical Atlas of Municipalities, Ministry of Economy and Finance (MEF)

Reference: https://asc.istat.it/ASC/

https://www.agenziaentrate.gov.it/portale/dichiarazioni-2021-archivio

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: The data for the year 2021 were obtained through the aggregation of the municipal level.

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Retirees With Low Pension Income (BE_3)

Name: Retirees with low pension income

Code: BE_3

Domain: Economic well-being (BE)

Description: Percentage of retirees who receive a gross monthly pension income of less than 500 euros on the total number of retirees.

Source: Istat/BES, Statistics of social security and assistance

Reference: https://www.istat.it/wp-content/uploads/2024/11/4.pdf

https://www.istat.it/scheda-qualita/statistiche-della-previdenza-e-dellassistenza-sociale-i-trattamenti-pensionistici/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Default Rate Of Bank Loans To Households (BE_4)

Name: Default rate of bank loans to households

Code: BE_4

Domain: Economic well-being (BE)

Description: Percentage ratio between the amounts of new bad loans in the year (loans to subjects declared insolvent or difficult to recover during the year) and the stock of non-bad loans in the year.

Source: Istat/BES, Bank of Italy - Central Credit Register

Reference: https://www.istat.it/wp-content/uploads/2024/04/4.pdf

https://www.bancaditalia.it/statistiche/index.html?com.dotmarketing.htmlpage.language=1

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Nominal Labour Productivity Per Hour Worked (BE_5)

Name: Nominal labour productivity per hour worked

Code: BE_5

Domain: Economic well-being (BE)

Description: This indicator represents the GDP per hour worked. Hours worked include total hours worked by all persons engaged in production in the domestic concept, i.e., by employees and self-employed persons, in either primary or secondary activity, engaged in a productive activity for a resident unit and receiving remuneration regardless of their place of residence. This ratio, expressed in euros per hour worked, indicates how much economic production activity in nominal prices in a given period can be attributed to each hour worked in the economy. Nominal labour productivity per hour worked is usually defined as a value of output (value added) produced in one hour of work. Nominal labour productivity per hour worked is calculated by dividing "GDP in current prices" by "hours worked".

Source: JRC/ REGIO - ARDECO

Reference: https://urban.jrc.ec.europa.eu/dataset/ARDECO-SUVGDH/metadata?lng=en

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain Homepage

INNOVATION, RESEARCH AND CREATIVITY (I)

This domain includes indicators that capture the capacity for innovation, scientific research, and cultural production, highlighting the role of knowledge and creativity in sustainable development for 107 Italian provinces.

The variables considered for this domain are:

- Employees in cultural enterprises (I_1)
- Patent propensity (I_2)
- Hours worked by industry (IC sector) (I_3)
- FTTH (Fiber to the home) (I_4)
- DSL (Digital Subscriber Line) (I_5)

Homepage

Employees In Cultural Enterprises (I_1)

Name: Employees in cultural enterprises

Code: I_1

Domain: Innovation, research and creativity (I)

Description: Percentage of employees in the local units of companies that carry out a cultural economic activity out of the total number of employees in the local units of companies. Eurostat defines the following economic activities as "totally cultural": publication of books, newspapers, magazines and periodicals, computer games; production of films, videos and television programs, sound recording and music publishing activities; programming and broadcasting activities; activities of news agencies; architectural activities; specialized design activities; creative, artistic and entertainment activities; libraries, archives, museums and other cultural activities.

Source: Istat/BES, Statistical Register of Local Units (ASIA UL)

Reference: https://www.istat.it/wp-content/uploads/2024/11/11.pdf

https://www.istat.it/scheda-qualita/registro-statistico-delle-unita-locali-asia-ul/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Patent Propensity (I_2)

Name: Patent propensity

Code: I_2

Domain: Innovation, research and creativity (I)

Description: Number of patent applications filed to the European Patent Office (EPO) per million of inhabitants.

Source: Istat/BES, Italian Patent and Trademark Office

Reference: https://www.istat.it/wp-content/uploads/2024/11/11.pdf

https://uibm.mise.gov.it/index.php/en/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: For missing units Bayesian Imputation Method (Panzera et al. 2016) has been used as explained in the SCI-WEB Deliverable D2.2.

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Hours Worked By Industry (IC sector) (I_3)

Name: Hours Worked by industry - IC sector

Code: I_3

Domain: Innovation, research and creativity (I)

Description: Share of hours actually worked by employees and self-employed persons in the IC sector on the hours actually worked by employees and self-employed in all sectors. The IC sector refers to information and communication activities.

Source: JRC / REGIO - ARDECO

Reference: https://urban.jrc.ec.europa.eu/dataset/ARDECO-RNLHZ/metadata?lng=en

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

FTTH (Fiber to the home) (I_4)

Name: FTTH (Fiber to the Home)

Code: I_4

Domain: Innovation, research and creativity (I)

Description: The term FTTH (Fiber to the Home) indicates all those connections in which the optical fiber, starting from the exchange, reaches the user's home. With this technology it is possible to reach and exceed the performance of 1Gbps. The coverage is expressed as a share on the total number of households and it is based on information relating to households location obtained from the EGON database. This database is widely used by electronic communications operators to detail the location of homes.

Source: AGCOM (Communications Guarantee Authority)

Reference:
https://agcom.maps.arcgis.com/apps/dashboards/770a9ca1c3684acc9bff5df3f2150854

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

DSL (Digital Subscriber Line) (I_5)

Name: DSL (Digital Subscriber Line)

Code: I_5

Domain: Innovation, research and creativity (I)

Description: DSL (Digital Subscriber Line) is the basic technology used to provide broadband over conventional telephone lines. The types of DSL used for standard fixed broadband (mainly ADSL or ADSL2+) deliver download speeds of at least 2 Mbps. A household has DSL coverage if it is a telephone exchange area fully enabled for DSL. The estimate of the share of overall families is based on information relating to the location of street numbers, obtained from the EGON database. This database is widely used by electronic communications operators to detail the location of homes.

Source: AGCOM (Communications Guarantee Authority)

Reference:
https://agcom.maps.arcgis.com/apps/dashboards/770a9ca1c3684acc9bff5df3f2150854

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

EDUCATION AND TRAINING (IS)

The indicators of the Education and Training domain at province level aim to assess access to and quality of education, lifelong learning, and the development of skills essential for personal and professional growth.

The variables considered for this domain are:

- Children who benefited from municipal children's services (IS_1)
- Participation in the school system of children aged 4-5 (IS_2)
- Graduates (IS_3)
- University enrollees (IS_4)
- Female graduates (IS_5)

Homepage

Children Who Benefited From Municipal Children's Services (IS_1)

Name: Children who benefited from municipal children's services

Code: IS_1

Domain: Education and training (IS)

Description: Percentage of children aged 0-2 who have used childcare services offered by public structures owned by the municipality or private structures. The services included are nursery schools, and supplementary services for early childhood.

Source: Istat/BES, Census survey on interventions and social services offered by individual or associated municipalities

Reference: https://www.istat.it/wp-content/uploads/2024/11/2.pdf

https://www.istat.it/scheda-qualita/indagine-sugli-interventi-e-i-servizi-sociali-dei-comuni-singoli-e-associati/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Participation In The School System Of Children Aged 4-5 (IS_2)

Name: Participation in the school system of children aged 4-5

Code: IS_2

Domain: Education and training (IS)

Description: Percentage of children aged 4-5 years participating in pre-primary education or in primary education on total children aged 4-5 years.

Source: Istat/BES, Processing of data from Ministry of Education

Reference: https://www.istat.it/wp-content/uploads/2024/11/2.pdf

https://www.mim.gov.it/dati-e-statistiche

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Graduates (IS_3)

Name: Graduates

Code: IS_3

Domain: Education and training (IS)

Description: Percentage of the new graduates out the total population aged between 22 and 25.

Source: MUR - Ministry of University and Research

Reference: https://dati-ustat.mur.gov.it/dataset/laureati/resource/0ac215ca-6cd2-4935-bf0e-14c66a13871f

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: For missing units Bayesian Imputation Method (Panzera et al. 2016) has been used as explained in the SCI-WEB Deliverable D2.2.

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

University Enrolees (IS_4)

Name: University enrolees

Code: IS_4

Domain: Education and training (IS)

Description: Percentage of university enrollers by province of residence out the total high school graduates of the same year.

Source: MUR - Ministry of University and Research

Reference: https://dati-ustat.mur.gov.it/dataset/immatricolati/resource/81c9d10d-5476-40bb-af1d-cc5d33cb64e7

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Female Graduates (IS_5)

Name: Female graduates

Code: IS_5

Domain: Education and training (IS)

Description: Percentage of new female graduates out the total population aged between 22 and 25.

Source: MUR - Ministry of University and Research

Reference: https://dati-ustat.mur.gov.it/dataset/laureati/resource/0ac215ca-6cd2-4935-bf0e-14c66a13871f

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: For missing units Bayesian Imputation Method (Panzera et al. 2016) has been used as explained in the SCI-WEB Deliverable D2.2.

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

WORK AND LIFE BALANCE (L)

This domain measures participation in the labour market, quality of employment, and the balance between work, personal, and family life for 107 Italian provinces.

The variables considered for this domain are:

- Gross hourly wage per hour paid of employee jobs (L_1)
- Hours worked per capita (L_2)
- Employment per capita (L_3)
- Gross hourly wage per hour paid of employee jobs for female (L_4)
- Unemployment (L_5)

Homepage

Gross Hourly Wage Per Hour Paid Of Employee Jobs (L_1)

Name: Gross hourly wage per hour paid of employee jobs

Code: L_1

Domain: Work and life balance (L)

Description: The average hourly wages is calculated for each employee job. It is obtained as the ratio between the gross wages and the estimated hours paid by the employer for each employee job. The variable is expressed in euros.

Source: Istat - Elaborations on INPS data - Observatory on Employees

Reference: https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1,Z0500LAB,1.0/LAB_EMPLWAGE/DCSC_RACLI/DCSC_RACLI_DISTRPROV/IT1,533_957_DF_DCSC_RACLI_8,1.0

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Hours Worked Per Capita (L_2)

Name: Hours worked per capita

Code: L_2

Domain: Work and life balance (L)

Description: This indicator represents the number of hours worked per capita. Hours worked per capita is calculated by dividing hours worked by employed persons on the annual average population.

Source: JRC / REGIO - ARDECO

Reference: https://urban.jrc.ec.europa.eu/dataset/ARDECO-RNLHTP/metadata?lng=en

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Employment Per Capita (L_3)

Name: Employment per capita

Code: L_3

Domain: Work and life balance (L)

Description: This indicator represents the number of jobs per capita. Employment per capita is calculated by dividing Total Employment (workplace based, employed persons) by annual average population.

Source: JRC / REGIO - ARDECO

Reference: https://urban.jrc.ec.europa.eu/dataset/ARDECO-SNETDP/metadata?lng=en

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Gross Hourly Wage Per Hour Paid Of Employee Jobs For Female (L_4)

Name: Gross hourly wage per hour paid of employee jobs for female

Code: L_4

Domain: Work and life balance (L)

Description: The average hourly wage, expressed in euros, is calculated for each female employee job. It is obtained as the ratio between the gross wages and the estimated hours paid by the employer for each employee job.

Source: Istat - Elaborations on INPS data - Observatory on Employees

Reference: https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1,Z0500LAB,1.0/LAB_EMPLWAGE/DCSC_RACLI/DCSC_RACLI_DISTRPROV/IT1,533_957_DF_DCSC_RACLI_8,1.0

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Unemployment (L_5)

Name: Unemployment

Code: L_5

Domain: Work and life balance (L)

Description: Unemployment rates represent unemployed persons as a percentage of the labour force.

Source: Istat, Eurostat

Reference:
https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1,Z0500LAB,1.0/LAB_OFFER/LAB_OFF_UNEMPLOY/DCCV_TAXDISOCCU1/IT1,151_914_DF_DCCV_TAXDISOCCU1_8,1.0

https://ec.europa.eu/eurostat/databrowser/view/tps00203/default/table?lang=en

Spatial Coverage: NUTS3 Regions of Italy (provinces)

Temporal Coverage: 8 years (2014-2021)

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: This variable was originally available only at NUTS2 level and has been disaggregated using Bayesian Imputation Method (BIM; Panzera and Postiglione, 2014), as explained in the SCI-WEB Deliverable D2.2.

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

LANDSCAPE AND CULTURAL HERITAGE (PA)

The indicators in this domain evaluate at the provincial level the level of protection, enhancement, and accessibility of cultural heritage and landscapes as key elements of collective identity and wellbeing.

The variables considered for this domain are:

- Density of Historic Parks/Gardens (PA_1)
- Spread of agritourism farms (PA_2)
- Density and importance of musems' heritage (PA_3)
- Capacity of hospitality establishments (PA_4)

Homepage

Density Of Historic Parks/Gardens (PA_1)

Name: Density of Historic Parks/Gardens

Code: PA_1

Domain: Landscape and cultural heritage (PA)

Description: Percentage ratio of the area of parks and gardens classified as "historic" and/or "of a significant public interest" by the Legislative Decree no. 42/2004 to the total area of the provincial capital Municipalities, normalized per 100 sq.km of urbanized surface (towns and inhabited areas).

Source: Istat/BES, Processing of data from Survey on urban environmental data and Census Mapping

Reference: https://www.istat.it/wp-content/uploads/2024/11/9.pdf

https://www.istat.it/informazioni-sulla-rilevazione/dati-ambientali/

https://www.istat.it/notizia/basi-territoriali-e-variabili-censuarie/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Spread Of Agritourism Farms (PA_2)

Name: Spread of agritourism farms

Code: PA_2

Domain: Landscape and cultural heritage (PA)

Description: Number of farmhouses per 100 sq.kms.

Source: Istat/BES, Survey on farmhouses

Reference: https://www.istat.it/wp-content/uploads/2024/11/9.pdf

https://www.istat.it/informazioni-sulla-rilevazione/rilevazione-sullagriturismo-informazioni-sulla-rilevazione-anno-2016-2/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Density And Importance Of Musems' Heritage (PA_3)

Name: Density and importance of musems' heritage

Code: PA_3

Domain: Landscape and cultural heritage (PA)

Description: Number of permanent exhibition facilities per 100 sq.km (museums, archaeological sites, and monuments open to public), weighed by the number of visitors. The weight of each facility is equal to V_i / (V/M), where V_i is the number of visitors to the facility, M is the total number of facilities and V is the total number of visitors.

Source: Istat - Survey on museums and other cultural institutions

Reference: https://www.istat.it/wp-content/uploads/2024/11/9.pdf

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain Homepage

Capacity Of Hospitality Establishments (PA_4)

Name: Capacity of hospitality establishments

Code: PA_4

Domain: Landscape and cultural heritage (PA)

Description: The survey quantifies, at the level of a single municipality, the number of establishments, beds, rooms and bathrooms for hotel facilities; of exercises and beds for the other structures. The units of analysis are the accommodation facilities present on the national territory, referring to each Italian municipality, classified according to national legislation and regional regulations and divided into: hotel facilities; hotels classified into five categories, distinguished by number of stars and tourist-hotel residences; non-hotel structures: campsites and areas equipped for campers and caravans, tourist villages, mixed forms of campsites and tourist villages, rental accommodation managed in a business manner, farmhouses, youth hostels, holiday homes, mountain refuges, other accommodation establishments not elsewhere classified, bed and breakfasts and other private accommodation. Furthermore, the variable is expressed per 100 sq.kms.

Source: Istat/BES, Statistical Atlas of Municipalities

Reference: https://www.istat.it/wp-content/uploads/2024/11/9.pdf

https://aster.istat.it/#/en/ast/categories/40/191/IT1,DF_TURISMO_CAP,1.0

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

POLITICS AND ISTITUTIONS (PO)

This domain explores trust in institutions, civic participation, and the quality of democratic life, focusing on citizens' engagement and institutional effectiveness in 107 provinces of Italy.

The variables considered for this domain are:

- Prison density (PO_1)
- Municipal administrators under 40 years old (PO_2)
- Female municipal administrators (PO_3)
- Municipalities: collection capacity (PO_4)

Homepage

Prison Density (PO_1)

Name: Prison density

Code: PO_1

Domain: Politics and institutions (PO)

Description: Percentage of prisoners in penal institutions on the total capacity of penal institutions.

Source: Istat/BES, Processing of data from the Ministry of Justice, Penitentiary Administration Department

Reference: https://www.istat.it/wp-content/uploads/2024/11/6.pdf

https://www.giustizia.it/giustizia/it/mg_1_14.page?selectedNode=0_2

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For the provinces of Macerata, Savona and Carbonia-Iglesias, the respective penitentiary institutions were closed. Consequently, missing data were imputed with a value of zero, since the prison population was moved to other provinces.

For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Municipal Administrators Under 40 Years Old (PO_2)

Name: Municipal administrators under 40 years old

Code: PO_2

Domain: Politics and institutions (PO)

Description: Percentage of administrators under the age of 40 on the total municipal administrators appointed.

Source: Istat/BES, Processed on data from the Ministry of the Interior - Register of local administrators

Reference: https://www.istat.it/wp-content/uploads/2024/11/6.pdf

https://dait.interno.gov.it/elezioni/anagrafe-amministratori

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Female Municipal Administrators (PO_3)

Name: Female municipal administrators

Code: PO_3

Domain: Politics and institutions (PO)

Description: Percentage of women on the total municipal administrators appointed.

Source: Istat/BES, Processed on data from the Ministry of the Interior - Register of local administrators

Reference: https://www.istat.it/wp-content/uploads/2024/11/6.pdf

https://dait.interno.gov.it/elezioni/anagrafe-amministratori

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Municipalities Collection Capacity (PO_4)

Name: Municipalities collection capacity

Code: PO_4

Domain: Politics and institutions (PO)

Description: Percentage ratio between the amount of accrual collections and the ascertained revenue.

Source: Istat/BES, Elaboration on Ministry of the Interior data

Reference: https://www.istat.it/wp-content/uploads/2024/11/6.pdf

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

QUALITY OF SERVICES (Q)

The Quality of Services domain focuses on accessibility, efficiency, and satisfaction related to public services, such as transport, waste management, and utilities.

The variables considered for this domain are:

- Hospital beds in high-care wards (Q_1)
- Seat-Km of public transport networks (Q_2)
- Hospital patient emigration to a different region (Q_3)
- Irregularities in electric power distribution (Q_4)
- Separate collection service for municipal waste (Q_5)

Homepage

Hospital Beds In High-Care Wards (Q_1)

Name: Hospital beds in high-care wards

Code: Q_1

Domain: Quality of services (Q)

Description: Hospital beds in high care wards in ordinary inpatient care in public and private health care institutions per 10,000 inhabitants.

Source: Istat/BES, Processing of Ministry of Health data

Reference: https://www.istat.it/wp-content/uploads/2024/11/12.pdf

https://www.salute.gov.it/portale/documentazione/p6_2_8.jsp?lingua=italiano

https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1,Z0810HEA,1.0/HEA_SERVICES/DCIS_OSPED/IT1,43_967_DF_DCIS_OSPED_2,1.0

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: The missing data for the province of Gorizia were directly imputed using the Istat dataset (IstatData), which provided information on ordinary places with high assistance for the period 2014-2016. The quantity of places was subsequently verified and confirmed through the regional budget documents.

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Seat-Km Of Public Transport Networks (Q_2)

Name: Seat-Km of public transport networks

Code: Q_2

Domain: Quality of services (Q)

Description: Seat-Km of public transport networks per capita in provincial capital municipalities.

Source: Istat/BES, Urban environmental data Survey

Reference: https://www.istat.it/wp-content/uploads/2024/11/12.pdf

https://www.istat.it/informazioni-sulla-rilevazione/dati-ambientali/

Spatial Coverage: NUTS3 Regions of Italy (provinces)

Temporal Coverage: 8 years (2014-2021)

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Hospital Patient Emigration To A Different Region (Q_3)

Name: Hospital patient emigration to a different region

Code: Q_3

Domain: Quality of services (Q)

Description: Percentage of hospital discharges carried out in regions other than that of residence on the total discharges of residents in the region. Data refer only to inpatient discharges for "acute" care (excluding hospitalizations of spinal unit, functional rehabilitation, neurorehabilitation, and long-term care).

Source: Istat/BES, Processing of Ministry of Health data

Reference: https://www.istat.it/wp-content/uploads/2024/11/12.pdf

https://www.salute.gov.it/portale/temi/p2_2.html

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: For missing units Bayesian Imputation Method (Panzera et al. 2016) has been used as explained in the SCI-WEB Deliverable D2.2.

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Irregularities In Electric Power Distribution (Q_4)

Name: Irregularities in electric power distribution

Code: Q_4

Domain: Quality of services (Q)

Description: Accidental long lasting electric power cuts (cuts without notice longer than 3 minutes) in average number per consumer.

Source: Istat/BES, Processing of data from the Authority for Electricity and Gas

Reference: https://www.istat.it/wp-content/uploads/2024/11/12.pdf

https://www.istat.it/storage/politiche-sviluppo/Energia.xls

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Separate Collection Service For Municipal Waste (Q_5)

Name: Separate collection service for municipal waste

Code: Q_5

Domain: Quality of services (Q)

Description: Percentage of resident population in municipalities with separate collection greater than or equal to 65%.

Source: Istat/BES, ISPRA

Reference: https://www.istat.it/wp-content/uploads/2024/11/12.pdf

https://indicatoriambientali.isprambiente.it/it/rifiuti/quantita-di-rifiuti-urbani-raccolti-modo-differenziato

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

SOCIAL RELATIONSHIPS (R)

Indicators in the Social Relationship domain at provincial level examine the strength and quality of social relationships, including networks of support, trust, and social participation.

The variables considered for this domain are:

- Net migration (R_1)
- Nonprofit organisations (R_2)
- Meetings with friends (R_3)
- Marriages (number of religious and civil marriages) (R_4)

Homepage

Net Migration (R_1)

Name: Net migration

Code: R_1

Domain: Social relationships (R)

Description: Net migration is the difference between the number of immigrants and the number of emigrants. In the context of the annual demographic balance, Eurostat produces net migration figures by taking the difference between total population change and natural change; this concept is referred to as 'net migration plus statistical adjustment'. The statistics on 'net migration plus statistical adjustment' are therefore affected by all the statistical inaccuracies in the two components of this equation, especially population change. From one country to another 'net migration plus statistical adjustment' may cover, besides the difference between inward and outward migration, other changes observed in the population figures between 1 January in two consecutive years which cannot be attributed to births, deaths, immigration and emigration.

Source: JRC / REGIO - ARDECO

Reference: https://urban.jrc.ec.europa.eu/dataset/ARDECO-SNMTN/metadata?lng=en

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Nonprofit Organisations (R_2)

Name: Nonprofit organisations

Code: R_2

Domain: Social relationships (R)

Description: Number of no-profit organizations per 10,000 inhabitants.

Source: Istat/BES, Statistical register and Census of nonprofit institutions

Reference: https://www.istat.it/wp-content/uploads/2024/11/5.pdf

https://www.istat.it/statistiche-per-temi/censimenti/istituzioni-non-profit/risultati/

https://www.istat.it/scheda-qualita/registro-delle-istituzioni-non-profit/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: Data on non-profits were not surveyed between 2012 and 2014. Therefore, for the year 2014, data from the closest available year—2011—were used.

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Meetings With Friends (R_3)

Name: Meetings with friends

Code: R_3

Domain: Social relationships (R)

Description: Thousands of people. The figure is calculated considering the share of individuals aged over 6 who, in their free time, that declare that they never meet friends or do not have any.

Source: Istat

Reference: https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1,Z0850DAI,1.0/SPORT_FRIENDS/IT1,83_63_DF_DCCV_AVQ_PERSONE_107,1.0

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: This variable was originally available only at NUTS2 level and has been disaggregated using Bayesian Imputation Method (BIM; Panzera and Postiglione, 2014), as explained in the SCI-WEB Deliverable D2.2.

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Marriages (Number Of Religious And Civil Marriages) (R_4)

Name: Marriages (number of religious and civil marriages)

Code: R_4

Domain: Social relationships (R)

Description: The number of religious and civil marriages per 10,000 inhabitants.

Source: Istat

Reference: https://esploradati.istat.it/databrowser/#/en/dw/categories/IT1,POP,1.0/POP_MARUNION/DCIS_MATRDEMO/IT1,24_186_DF_DCIS_MATRDEMO_9,1.0

Spatial Coverage: NUTS3 Regions of Italy (provinces)

Temporal Coverage: 8 years (2014-2021)

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: For Sardinia, the variable was modified to account for changes in provincial boundaries and was proportionally redistributed based on the population totals of both the former (110 provinces) and current (107 provinces) administrative divisions.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

HEALTH (H)

The indicators in the Health domain monitor the physical health status of the population, the availability and quality of healthcare services, and health-related behaviors for 107 Italian provinces.

The variables considered for this domain are:

- Avoidable Mortality (age 0-74) (H_1)
- Life expectancy at birth (H_2)
- Road accidents mortality rate (15-34 years old) (H_3)
- Age-standardised cancer mortality rate (20-64 years old) (H_4)
- Infant mortality rate (H_5)

Homepage

Avoidable Mortality (age 0-74) (H_1)

Name: Avoidable Mortality (age 0-74)

Code: H_1

Domain: Health (H)

Description: Deaths of persons aged 0-74, due to causes identified as treatable (in the light of medical knowledge and technology at the time of death, most deaths from that cause could be avoided through optimal quality health care) or preventable (in the light of understanding of the determinants of health at the time of death, most deaths from that cause could be avoided by public health interventions in the broadest sense). The definition of the lists of treatable and preventable causes of mortality is based on a joint OECD/Eurostat work, revised in November 2019. Standardized rates with population aged 0-74 per 10,000 residents.

Source: Istat/BES, Vital register on deaths and causes of death

Reference: https://www.istat.it/wp-content/uploads/2024/11/1.pdf

https://www.istat.it/informazioni-sulla-rilevazione/rilevazione-sulle-cause-di-morte/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain Homepage

Life Expectancy At Birth (H_2)

Name: Life expectancy at birth

Code: H_2

Domain: Health (H)

Description: Life expectancy expresses the average number of years that a child born in a given calendar year can expect to live if exposed during his whole life to the risks of death observed in the same year at different ages.

Source: Istat/BES, Life tables of Italian population

Reference: https://www.istat.it/wp-content/uploads/2024/11/1.pdf

https://demo.istat.it/app/?i=TVM&a=2023&l=en

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Road Accidents Mortality Rate (15-34 years old) (H_3)

Name: Road accidents mortality rate (15-34 years old)

Code: H_3

Domain: Health (H)

Description: Mortality rate in road accidents by five year age groups for people aged 15-34 years per 10,000 inhabitants.

Source: Istat/BES. As regards deaths: Survey of road accidents with injuries to people. As regards population: Survey on the resident municipal population broken down by sex, year of birth and marital status.

Reference: https://www.istat.it/wp-content/uploads/2024/11/1.pdf

https://www.istat.it/informazioni-sulla-rilevazione/incidenti-stradali/

https://www.istat.it/scheda-qualita/popolazione-residente-comunale-per-sesso-anno-di-nascita-e-stato-civile/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: For missing units Bayesian Imputation Method (Panzera et al. 2016) has been used as explained in the SCI-WEB Deliverable D2.2.

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Age-Standardised Cancer Mortality Rate (20-64 years old) (H_4)

Name: Age-standardised cancer mortality rate (20-64 years old)

Code: H_4

Domain: Health (H)

Description: Mortality rate for cancer (initial cause) by five-year age groups for people aged 20-64 years per 10,000 inhabitants.

Source: Istat/BES. As regards deaths: Survey on deaths and causes of death. As regards population: Survey on the resident municipal population broken down by sex, year of birth and marital status.

Reference: https://www.istat.it/wp-content/uploads/2024/11/1.pdf

https://www.istat.it/informazioni-sulla-rilevazione/rilevazione-sulle-cause-di-morte/

https://demo.istat.it/app/?i=POS&a=2024&l=en

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic Well-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Infant Mortality Rate (H_5)

Name: Infant mortality rate

Code: H_5

Domain: Health (H)

Description: Deaths during the first year of life per 10.000 children born alive.

Source: Istat/BES. For deaths: Vital register on deaths and causes of death. For live births: Migration and calculation of yearly resident population.

Reference: https://www.istat.it/wp-content/uploads/2024/11/1.pdf

https://www.istat.it/informazioni-sulla-rilevazione/rilevazione-sulle-cause-di-morte/

https://siqual.istat.it/SIQual/visualizza.do?id=5000088

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: For missing units Bayesian Imputation Method (Panzera et al. 2016) has been used as explained in the SCI-WEB Deliverable D2.2.

Notes: -

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

SAFETY (S)

The Safety domain assesses the perception and experience of safety in everyday life, including crime rates, road safety, and other risks affecting personal security for 107 provinces in Italy.

The variables considered for this domain are:

- Pick-pocketing (S_1)
- Burglary (S_2)
- Robbery (S_3)
- Report crimes to the police (S_4)
- Reports of sexual violence (S_5)

Homepage

Pick-Pocketing (S_1)

Name: Pickpocketing

Code: S_1

Domain: Safety (S)

Description: Victims of pickpocketing per 100,000 inhabitants: the number of victims is calculated using data on victims who reported pickpocketing to the police, adjusted by the number of non-reporting victims from the Citizens' Safety Survey, using a specific correction factor by geographical area and by sex and age group.

Source: Istat/BES, Processing of data from crimes reported to Police Forces (Ministry of Interiors) and data on Citizens 'Safety Survey

Reference: https://www.istat.it/wp-content/uploads/2024/11/7.pdf

https://ucs.interno.gov.it/ucs/contenuti/Numero_dei_delitti_denunciati_all_autorit_agrave_giudiziaria_dalle_forze_di_polizia_int_00062-7730889.htm

https://www.istat.it/en/press-release/safety-perception/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: To reconstruct the values for the province of South Sardinia, it was noted that this area largely fell under the territorial jurisdiction of the Cagliari prefecture. Consequently, the data were proportionally redistributed between South Sardinia and Cagliari based on their respective annual population densities.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Burglary (S_2)

Name: Burglary

Code: S_2

Domain: Safety (S)

Description: Victims of burglaries per 100,000 households: the number of victims is calculated using data on victims who reported burglary to the police, adjusted by the number of nonreporting victims from the Citizens' Safety Survey, using a specific correction factor by geographical area.

Source: Istat/BES, Processing of data from crimes reported to Police Forces (Ministry of Interiors) and data on Citizens' Safety Survey

Reference: https://www.istat.it/wp-content/uploads/2024/11/7.pdf

https://ucs.interno.gov.it/ucs/contenuti/Numero_dei_delitti_denunciati_all_autorit_agrave_giudiziaria_dalle_forze_di_polizia_int_00062-7730889.htm

https://www.istat.it/en/press-release/safety-perception/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: To reconstruct the values for the province of South Sardinia, it was noted that this area largely fell under the territorial jurisdiction of the Cagliari prefecture. Consequently, the data were proportionally redistributed between South Sardinia and Cagliari based on their respective annual population densities.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Robbery (S_3)

Name: Robbery

Code: S_3

Domain: Safety (S)

Description: Victims of robberies per 100,000 inhabitants: the number of victims is calculated using data on victims who reported robbery to the police, adjusted by the number of non-reporting victims from the Citizens' Safety Survey, using a specific correction factor by geographical area and by sex and age group.

Source: Istat/BES, Processing of data from crimes reported to Police Forces (Ministry of Interiors) and data on citizens 'Safety Survey

Reference: https://www.istat.it/wp-content/uploads/2024/11/7.pdf

https://ucs.interno.gov.it/ucs/contenuti/Numero_dei_delitti_denunciati_all_autorit_agrave_giudiziaria_dalle_forze_di_polizia_int_00062-7730889.htm

https://www.istat.it/en/press-release/safety-perception/

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: To reconstruct the values for the province of South Sardinia, it was noted that this area largely fell under the territorial jurisdiction of the Cagliari prefecture. Consequently, the data were proportionally redistributed between South Sardinia and Cagliari based on their respective annual population densities.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Report Crimes To The Police (S_4)

Name: Report crimes to the police

Code: S_4

Domain: Safety (S)

Description: Crimes per 100,000 thousands inhabitants reported to the judicial authorities by the State Police, the Carabinieri and the Financial Police.

Source: Istat

Reference: http://dati.istat.it/Index.aspx?lang=en&SubSessionId=42194570-a793-4751-b85c-f39c4c82453d

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: To reconstruct the values for the province of South Sardinia, it was noted that this area largely fell under the territorial jurisdiction of the Cagliari prefecture. Consequently, the data were proportionally redistributed between South Sardinia and Cagliari based on their respective annual population densities.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Report Of Sexual Violence (S_5)

Name: Reports of sexual violence

Code: S_5

Domain: Safety (S)

Description: Number of sexual violence per 100,000 inhabitants reported by the police to the judicial authorities.

Source: Istat

Reference: http://dati.istat.it/Index.aspx?lang=en&SubSessionId=68681521-23ca-4988-85cb-e626339289fe

Spatial Coverage: NUTS3 Regions of Italy (provinces).

Temporal Coverage: 8 years (2014-2021).

Format: .csv

Classification: NUTS 2021 (IT) https://ec.europa.eu/eurostat/web/nuts

Integration: -

Notes: To reconstruct the values for the province of South Sardinia, it was noted that this area largely fell under the territorial jurisdiction of the Cagliari prefecture. Consequently, the data were proportionally redistributed between South Sardinia and Cagliari based on their respective annual population densities.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

3. Metadata At Municipal Level

These metadata refer to the 25 indicators provided by SCI-WEB at the municipality level for the years 2018 and 2021. They complement the two datasets at the municipal level provided in the SCI-WEB project. Please note that the two datasets correspond to different municipal partitions, based on ISTAT's definitions of administrative boundaries of municipalities for 2018 and 2021 (see, ISTAT, 2025).

The domains considered are:

- Environment (a)
- Economic well-being (be)
- Innovation, research and creativity (i)
- Education and training (is)
- Work and life balance (l)
- Landscape and cultural heritage (pa)
- Politics and istitutions (po)
- Quality of services (q)
- Social relationships (r)
- Health (h)
- Safety (s)

ENVIRONMENT (a)

At the municipality level, the indicators of the Environment domain aim at monitoring the evolution local communities well-being with respect to the natural environment they live.

The variables considered for this domain are:

- Municipal waste generated (a_1)
- Protected natural areas (a_2)
- Areas at risk of landslides (a_3)
- Motorisation rate with high emissions (a_4)

Homepage

Municipal Waste Generated (a_1)

Name: Municipal waste generated

Code: a_1

Domain: Environment (a)

Description: Total municipal waste generated per capita (in kgs).

Source: Ispra, Istat/BES

Reference: https://www.catasto-rifiuti.isprambiente.it/index.php?pg=findComune

https://www.istat.it/wp-content/uploads/2024/11/10.pdf

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: -

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Protected Natural Areas (a_2)

Name: Protected natural areas

Code: a_2

Domain: Environment (a)

Description: Percentage of territorial surface covered by protected terrestrial natural areas included in the Official List of Protected Areas (*EUAP*) or belonging to the Natura 2000 Network.

Source: Statistical Atlas of Municipalities (ASC), Ispra: processing of data from the Ministry of the Environment and Energy Security.

Reference:
https://aster.istat.it/#/en/atlante_comuni/categories/37/IFC_ASC/IT1,DF_COMP_FRA_IND_MUNICIPAL,1.0?t=3

https://www.isprambiente.gov.it/en?set_language=en

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: -

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Areas At Risk Of Landslides (a_3)

Name: Areas at risk of landslides

Code: a_3

Domain: Environment (a)

Description: Percentage of municipal area subject to high and very high landslide hazard (areas P3-P4 of the Hydrogeological Structure Plan).

Source: Statistical Atlas of Municipalities (ASC), Istat: processing of data Ispra and Istat

Reference:
https://aster.istat.it/#/en/atlante_comuni/categories/37/IFC_ASC/IT1,DF_COMP_FRA_IND_MUNICIPAL,1.0?t=3

https://www.isprambiente.gov.it/en?set_language=en

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: -

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Motorisation Rate With High Emissions (a_4)

Name: Motorisation rate with high emissions

Code: a_4

Domain: Environment (a)

Description: The indicator expresses the number of vehicles with the highest pollutant emissions (Euro classes 0 to 3) per 100 inhabitants.

Source: Statistical Atlas of Municipalities (ASC), Istat: processing of data ACI and Istat - Permanent Census on Population and Housing

Reference:
https://aster.istat.it/#/en/atlante_comuni/categories/37/IFC_ASC/IT1,DF_COMP_FRA_IND_MUNICIPAL,1.0?t=3

https://aci.gov.it/attivita-e-progetti/studi-e-ricerche/open-data/

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: -

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

ECONOMIC WELL-BEING (be)

The indicators of the Economic Well-being domain at municipality level aim at giving an overview about the evolution of the economic well-being of citizens.

The variables considered for this domain are:

- Income (Total income tax returns – IRPEF) (be_1)
- Retirement income (be_2)
- Number of local units of active enterprises (be_3)

Homepage

Income (Total Income Tax Returns – IRPEF) (be_1)

Name: Income (Total income tax returns - IRPEF)

Code: be_1

Domain: Economic well-being (be)

Description: The personal income tax (*Irpef*) is the tax payable by natural person for the possession of the following income sources: land, i.e. buildings and land; capital; employment (including income assimilated to that of employment and pension income); self-employment; business; various (listed in article 67 of the Consolidated Income Tax Act).

Natural persons resident and non-resident in the territory of the State are taxable subjects by Irpef. Income (total income tax return - IRPEF) is calculated by dividing the total income tax return - IRPEF by the average annual population of taxpayers per municipality. The variable is expressed in euros.

Source: Revenue Agency, Statistical Atlas of Municipalities (ASC), Ministry of Economy and Finance (*MEF*)

Reference: https://asc.istat.it/ASC/

https://www.agenziaentrate.gov.it/portale/dichiarazioni-2021-archivio

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Retirement Income (be_2)

Name: Retirement income

Code: be_2

Domain: Economic well-being (be)

Description: It represents the amounts received for the cessation of work activity or other reasons provided by law (e.g. disability pension, survivor's pension, social pension, voluntary contribution). The variable has been normalized on the number of pension income recipients, and it is expressed in euros.

Source: Revenue Agency, Ministry of Economy and Finance (MEF)

Reference: https://www.agenziaentrate.gov.it/portale/dichiarazioni-2021-archivio

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Number Of Local Units Of Active Enterprises (be_3)

Name: Number of local units of active enterprises

Code: be_3

Domain: Economic well-being (be)

Description: Number of active businesses in the territory. By businesses we only consider the active legal units. The variable is expressed as number of active units per 100 inhabitants.

Source: Statistical Atlas of Municipalities (ASC)

Reference: https://aster.istat.it/#/en/atlante_comuni/categories/25/158/400/IT1,183_285,1.0?t=3

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

INNOVATION, RESEARCH AND CREATIVITY (i)

This domain encompasses indicators that capture for each municipality its capacity for innovation and highlighting the role of knowledge and creativity in sustainable development.

The variables considered for this domain are:

- Persons employed in hi-tech and knowledge intensive enterprises (i_1)
- Internet speed (i_2)

Homepage

Persons Employed In Hi-Tech And Knowledge Intensive Enterprises (i_1)

Name: Persons employed in hi-tech and knowledge intensive enterprises

Code: i_1

Domain: Innovation, research and creativity (i)

Description: Percentage of individuals employed in high-tech and knowledge intensive activities on total people employed in non-agricultural enterprises.

Source: Statistical Atlas of Municipalities (ASC)

Reference: https://aster.istat.it/#/en/atlante_comuni/categories/25/158/SIR/SIR_OCC/IT1,DF_DIPS_SIR_IND_TERR_DRT_OCC,1.0?t=3

Spatial Coverage: Municipalities of Italy

Years: 2018 and 2021

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Internet Speed (i_2)

Name: Internet speed

Code: i_2

Domain: Innovation, research and creativity (i)

Description: Average download speed values expressed in megabit per second (Mbps).

Source: EDJNet (European Data Journalism Network)

Reference: https://datavis.europeandatajournalism.eu/obct/connectivity/

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The 2018 data correspond to the first quarter of 2019, while the 2021 data refer to the last quarter of that year.

The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

EDUCATION AND TRAINING (is)

The indicators of the Education and Training domain at municipality level allow us to assess the quality of education for each municipality of Italy with a focus on gender gaps.

The variables considered for this domain are:

- Graduates (is_1)
- Female graduates (is_2)

Homepage

Graduates (is_1)

Name: Graduates

Code: is_1

Domain: Education and training (is)

Description: Percentage new graduates on the total population.

Source: MUR - Ministry of University and Research

Reference: https://dati-ustat.mur.gov.it/dataset/laureati/resource/0ac215ca-6cd2-4935-bf0e-14c66a13871f

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: For missing units Bayesian Imputation Method (Panzera et al. 2016) has been used as explained in the SCI-WEB Deliverable D2.2.

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Female Graduates (is_2)

Name: Female graduates

Code: is_2

Domain: Education and training (is)

Description: Percentage of new female graduates on the total population.

Source: MUR - Ministry of University and Research

Reference: https://dati-ustat.mur.gov.it/dataset/laureati/resource/0ac215ca-6cd2-4935-bf0e-14c66a13871f

Spatial Coverage: Municipalities of Italy

Years: 2018 and 2021

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: For missing units Bayesian Imputation Method (Panzera et al. 2016) has been used as explained in the SCI-WEB Deliverable D2.2.

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

WORK AND LIFE BALANCE (*I*)

At municipality level, the Work and Life Balance domain measures participation in the labor market and with a focus on potential gender inequalities.

The variables considered for this domain are:

- Employment rate (20-64 years) (I_1)
- Female employees (I_2)
- Income from employment (I_3)

Homepage

Employment Rate (20-64 years) (*I_1*)

Name: Employment rate (20-64 years)

Code: *I_1*

Domain: Work and life balance (I)

Description: Percentage ratio between employed individuals aged 20-64 and the population aged 20-64.

Source: Statistical Atlas of Municipalities (ASC), Istat: processing of data from the Permanent Census of Population and Housing

Reference:
https://aster.istat.it/#/en/atlante_comuni/categories/37/IFC_ASC/IT1,DF_COMP_FRA_IND_MUNICIPAL,1.0?t=3

https://www.istat.it/en/statistical-themes/censuses/permanent-census-of-population-and-housing/results-of-the-permanent-population-census/

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Female Employees (*I*_2)

Name: Female employees

Code: *I*_2

Domain: Work and life balance (I)

Description: Percentage of female employees in non-agricultural enterprises on total employees in non-agricultural enterprises.

Source: Statistical Atlas of Municipalities (ASC)

Reference: https://aster.istat.it/#/en/atlante_comuni/categories/25/158/SIR/SIR_OCC/IT1,DF_DIPS_SIR_IND_TERR_DRT_OCC,1.0?t=3

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Income From Employment (*I*_3)

Name: Income from employment

Code: *I*_3

Domain: Work and life balance (I)

Description: Employment income is derived from the relationships involving the provision of work, with any qualification, under the direction of others, including homework when it is considered employment under the provisions of labor legislation. The variable has been normalized on the number of employees and it is expressed in euros.

Source: Revenue Agency, Ministry of Economy and Finance (MEF)

Reference: https://www.agenziaentrate.gov.it/portale/dichiarazioni-2021-archivio

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985)..

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

LANDSCAPE AND CULTURAL HERITAGE (pa)

The indicators in this domain evaluate the enhancement and accessibility of cultural heritage and local attractions as key elements of collective identity and wellbeing at municipality level.

The variables considered for this domain are:

- Capacity of collective tourist accomodation (pa_1)
- Spread of agritourism farms (pa_2)

Homepage

Capacity Of Collective Tourist Accommodation (pa_1)

Name: Capacity of collective tourist accomodation

Code: pa_1

Domain: Landscape and cultural heritage (pa)

Description: The variable quantifies, at the level of a single municipality, the number of establishments, beds, rooms and bathrooms for hotel facilities structures per sq.km. The units of analysis are the accommodation facilities present on the national territory, referring to each Italian municipality, classified according to national legislation and regional regulations and divided into: hotel facilities: hotels classified into five categories, distinguished by number of stars and tourist-hotel residences; non-hotel structures: campsites and areas equipped for campers and caravans, tourist villages, mixed forms of campsites and tourist villages, rental accommodation managed in a business manner, farmhouses, youth hostels, holiday homes, mountain refuges, other accommodation establishments not elsewhere classified, bed and breakfasts and other private accommodation.

Source: Statistical Atlas of Municipalities (ASC), Istat/BES

Reference:
https://aster.istat.it/#/en/atlante_comuni/categories/40/191/IT1,DF_TURISMO_CAP,1.0?t=3

https://www.istat.it/wp-content/uploads/2024/11/9.pdf

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Spread Of Agritourism Farms (pa_2)

Name: Spread of agritourism farms

Code: pa_2

Domain: Landscape and cultural heritage (pa)

Description: Spread of agritourism farms calculated as the number of farmhouses per sq.km.

Source: Istat/BES, Survey on rural tourism accommodations

Reference: https://aster.istat.it/ - /en/ast/categories/5/AGRITUR/IT1,101_1039,1.0?t=3

https://aster.istat.it/#/en/ast/categories/5/5_CENSAGR/IT1,DF_DCAT_CENSAGRIC2020_GAIN,1.0?t=3

https://www.istat.it/wp-content/uploads/2024/11/9.pdf

Spatial Coverage: Municipalities of Italy

Years: 2018 and 2021

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: -

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021. Data for were not available and, therefore, data from December 2020 were used instead.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

POLITICS AND ISTITUTIONS (po)

At municipality level, this domain explores trust civic participation and the quality of democratic life, focusing on citizens' engagement and institutional effectiveness.

The variables considered for this domain are:

- Female municipal administrators (po_1)
- Municipal administrators under 40 years old (po_2)

Homepage

Female Municipal Administrators (po_1)

Name: Female municipal administrators

Code: po_1

Domain: Politics and institutions (po)

Description: Female municipal administrators calculated as the percentage of women on the total municipal administrators appointed.

Source: Istat/BES, Processed on data from the Ministry of the Interior - Register of local administrators

Reference: https://www.istat.it/wp-content/uploads/2024/11/6.pdf

https://dait.interno.gov.it/elezioni/anagrafe-amministratori

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Municipal Administrators Under 40 Years Old (po_2)

Name: Municipal administrators under 40 years old

Code: po_2

Domain: Politics and institutions (po)

Description: Municipal administrators under 40 years old calculated as the percentage of administrators under the age of 40 on the total municipal administrators appointed.

Source: Istat/BES, Processed on data from the Ministry of the Interior - Register of local administrators

Reference: https://www.istat.it/wp-content/uploads/2024/11/6.pdf

https://dait.interno.gov.it/elezioni/anagrafe-amministratori

Spatial Coverage: Municipalities of Italy

Years: 2018 and 2021

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

QUALITY OF SERVICES (q)

The Quality of Services domain focuses on accessibility and efficiency of public services at the municipality level.

The variables considered for this domain are:

- Means of transport (q_1)
- Percentage of separate waste collection (q_2)
- Index of accessibility to essential services (q_3)

Homepage

Means Of Transport (q_1)

Name: Means of transport

Code: q_1

Domain: Quality of services (q)

Description: Number of buses and trolleybuses in the municipality per 1,000 inhabitants.

Source: Statistical Atlas of Municipalities (ASC), Automobile Club of Italy (ACI)

Reference: https://aster.istat.it/#/en/ast/categories/43/203/IT1,41_993,1.0

https://aci.gov.it/servizio/fornitura-dati-pra/

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: -

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Percentage Of Separate Waste Collection (q_2)

Name: Percentage of separate waste collection

Code: q_2

Domain: Quality of services (q)

Description: Percentage of separate waste collection on the total amount of waste generated for each municipality.

Source: Ispra

Reference: https://www.catasto-rifiuti.isprambiente.it/index.php?pg=findComune

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: -

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Index Of Accessibility To Essential Services (q_3)

Name: Index of accessibility to essential services

Code: q_3

Domain: Quality of services (q)

Description: The indicator measures the distance between the center (identified with the centroid of the census section where the municipal house is located) of a municipality and the corresponding nearest main center. Distance is expressed in terms of road travel times by car (in minutes). The main center municipalities are preliminarily identified based on the presence of three essential services: hospitals; complete cycle of secondary schools (at least one between classical and scientific high schools and at least one between technical and vocational schools); railway station with active passenger service, at least silver category.

Source: Statistical Atlas of Municipalities (ASC), Istat: processing of commercial road graph data (TOM-TOM) and Territorial Bases (Istat)

Reference:
https://aster.istat.it/#/en/atlante_comuni/categories/37/IFC_ASC/IT1,DF_COMP_FRA_IND_MUNICIPAL,1.0?t=3

https://www.istat.it/en/press-release/accessibility-of-municipalities-to-major-transport-infrastructures/

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

SOCIAL RELATIONSHIPS (r)

At municipality level, indicators on Social Relationships examine the strength and quality of social networks and its support.

The variables considered for this domain are:

- Marriages (r_1)
- Public spending on youth policies (r_2)

Homepage

Marriages (Number Of Religious And Civil Marriages) (r_1)

Name: Marriages

Code: r_1

Domain: Social relationships (r)

Description: Marriages represents the number of religious and civil marriages per 1,000 inhabitants.

Source: Istat

Reference: https://demo.istat.it/app/?i=MA1&a=2023&l=en

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

Public Spending On Youth Policies (r_2)

Name: Public spending on youth policies

Code: r_2

Domain: Social relationships (r)

Description: Share of the overall public spending on youth policies accounted by each municipality on the total amount of population. The spending was calculated as the sum of several items, including sports facilities; organization and participation in events and conferences; psychological, social, and religious assistance; accessories for sports and recreational activities; current government transfers to offices providing welfare, recreational and cultural services; sports facilities of cultural, historical and artistic value. The variables is expressed in euros.

Source: Ministry of Economy and Finance (MEF)

Reference: https://bdap-opendata.rgs.mef.gov.it/

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: -

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

HEALTH (h)

The indicators in the Health domain monitor the physical status of the population at municipality level. The variables considered for this domain are:

- Road accidents (h_1)

Homepage

Road Accidents (h_1)

Name: Road accidents

Code: h_1

Domain: Health (h)

Description: Number of deaths and injuries in road accidents per 1,000 inhabitants.

Source: Statistical Atlas of Municipalities (ASC)

Reference: https://aster.istat.it/#/en/ast/categories/43/209/IT1,41_983,1.0

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: Missing data were replaced using the weighted mean of values observed in first-order neighboring locations (Panzera et al. 2016; Haining, 2003; Tobler and Kennedy, 1985).

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

SAFETY (s)

This domain assesses the perception and experience of safety in everyday life for Italian municipalities. The variable considered for this domain is:

- Public spending on security (s_1)

Homepage

Public Spending On Security (s_1)

Name: Public spending on security

Code: s_1

Domain: Safety (s)

Description: Share of the overall public expenditure on safety accounted by each municipality on the total amount of population. The spending was calculated as the sum of various items, including safety services; ordinary maintenance and repair of transport vehicles for civil, safety and public order use; transport vehicles for civil, safety and public order; other materials for military, public order and safety purposes. The variable is expressed in euros.

Source: Ministry of Economy and Finance (MEF)

Reference: https://bdap-opendata.rgs.mef.gov.it/

Spatial Coverage: Municipalities of Italy.

Years: 2018 and 2021.

Format: .csv

Classification: LAU (IT) https://www.istat.it/notizia/confini-delle-unita-amministrative-a-fini-statistici-al-1-gennaio-2018-2/

Integration: -

Notes: The territorial configuration of Italian municipalities changed between the two reference years. Specifically, there were 7,960 municipalities in 2018, compared to 7,904 in 2021.

Project: Spatial Composite indicators and inequality Indices for measuring socio-economic WEll-Being (SCI-WEB) - Spoke 8 GRINS.

Domain
Homepage

4. Concluding remarks

The integration of multi-level datasets that capture both shared and context-specific variables constitutes a significant advancement in the study of territorial inequalities and spatial well-being. Grounded in the methodological inspiration of the SCI-WEB framework, these datasets provide a robust basis for conducting analyses using both standard statistical tools and spatially targeted methodologies (Panzera and Postiglione, 2020). They are particularly effective in identifying structural disparities and localized vulnerabilities that may be obscured at more aggregated levels of analysis.

The availability of granular data at levels finer than NUTS2, especially at the Local Administrative Unit (LAU) level, facilitates the construction of detailed spatial well-being indicators (Cartone and Postiglione, 2021). These indicators capture the complex, multidimensional nature of well-being—encompassing environmental sustainability, gender equality, and cultural dimensions—while also offering actionable insights for territorial governance. This analytical framework will be further enriched through its application to relevant territorial partitions, such as Inner Areas, Rural-Urban typologies, and the socio-economic divide between the Southern and Northern regions of the country. Such disaggregated perspectives will allow for more precise and differentiated policy responses tailored to the specific needs of each area (Barca et al. 2012).

Moreover, this type of analysis provides a valuable lens for supporting the objectives of the Recovery and Resilience Plan and Sustainable Development Goals, particularly in promoting inclusive growth, digital and green transitions, and reducing inequalities. By informing assessments of gender gaps, social cohesion, and environmental challenges—such as reducing ecological footprints and fostering circular economy principles—these data-driven tools align closely with the goals of the European Green Deal and broader EU cohesion policies.

In this context, the contributions of the GRINS project, and in particular Spoke 8, are especially timely. By embracing an integrated, data-informed approach to sustainability and equity, this work provides a solid foundation for more resilient, just, and ecologically responsible territorial development strategies.

Finally, this report informs scientific communities and other potential stakeholders on the advancement of territorial knowledge at local levels offering applied insights that can be accessed through the AMELIA platform for diverse research, planning, and policy purposes in line with the broader objectives of the GRINS project.

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